

**UNIVERSITY OF TRANSPORT HO CHI MINH
CITY
INSTITUTE OF INFORMATION TECHNOLOGY**



FINAL TERM REPORT

Software Technology Learning Course

PROJECT TOPIC

**PERSONALIZED CAREER
ORIENTATION
& LEARNING ROADMAP PLATFORM
FOR SOFTWARE ENGINEERING
STUDENTS**

Group 2

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Contents

I	Project Introduction	1
1	Overview	1
1.1	Project Information	1
1.2	Project Team	2
2	Product Background	2
3	Existing Systems	3
4	Business Opportunity	3
5	Software Product Vision	4
6	Project Scope & Limitations	5
6.1	Major Features	5
6.1.1	Major Features for Administrators	5
6.1.2	Major Features for Student	5
6.1.3	Major Features for Academic Counselor	6
6.1.4	Major Features for Industry Mentor	6
6.2	Limitations & Exclusions	6
II	Project Management Plan	8
1	Overview	8
1.1	Project Scope and Estimation	8
1.2	Testing Objectives	9
1.3	Project Risks	9
2	Management Approach	10
2.1	Development Methodology	10
2.2	Quality Management	10
2.3	Training Plan	10
3	Project Deliverables	11
4	Responsibility Assignments	11

5	Project Communications	12
6	Configuration Management	12
6.1	Document Management	12
6.2	Source Code Management	12
6.3	Version Control Policy	12
III Software Requirement Specification		14
1	Product Overview	14
2	User Requirements	14
3	Functional Requirements	14
3.1	System Functional Overview	14
3.2	Authentication	14
3.2.1	Login	14
3.3	User Management	16
3.3.1	Manage Users	17
3.3.2	Import Students	17
3.3.3	Import Mentors	18
3.4	Career Assessment Management	18
3.4.1	Student Take Career Assessment	19
3.4.2	View Assessment Result	19
3.4.3	Mentor Review Assessment	20
3.4.4	Manage Assessment Questions	21
3.5	Mentorship Management	21
3.5.1	Student Request Mentor	22
3.5.2	Mentor Review Request	22
3.5.3	Mentor Respond Request	23
3.5.4	Schedule Mentoring Session	24
3.5.5	View Mentoring History	24
3.6	Learning Roadmap Management	25
3.6.1	Generate Learning Roadmap	25
3.6.2	Update Learning Progress	26
3.6.3	Export Learning Roadmap	26
3.7	Learning Resource Management	27
3.7.1	Generate Learning Resource Recommendations	27
3.7.2	Manage Learning Resources	28

3.8	Timeline Management	29
3.8.1	Manage Career Development Timeline	29
3.9	Blog Management	31
3.9.1	Create Blog Post	31
3.9.2	Interact with Blog Post	33
3.9.3	Apply for Internship or Team Recruitment	35
3.10	Semester Management	37
3.10.1	Manage Semester	37
3.11	Criteria Form Management	39
3.11.1	Manage Criteria Form	39
4	Non-Functional Requirements	41
4.1	External Interfaces	41
4.2	Quality Attributes	42
4.2.1	Usability	42
4.2.2	Reliability	42
4.2.3	Performance	43
5	Requirement Appendix	43
5.1	Business Rules	44
5.2	Common Requirements	44
5.3	Application Messages List	45
5.4	Other Requirements	45

List of Figures

III.1 Login Interface	16
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List of Tables

I.1	Project Team	2
II.1	Work Breakdown Structure (WBS) and Project Effort Estimation . . .	8
II.2	Testing Objectives	9
II.3	Project Risks	9
II.4	Training Plan	10
II.5	Project Deliverables	11
II.6	Responsibility Assignment Matrix	11
II.7	Project Communications	12
III.1	Business Rules	44
III.2	Application Messages	45

I. Project Introduction

1. Overview

1.1 Project Information

- **Project name:** Personalized Career Orientation & Learning Roadmap Platform for Software Engineering Students
- **Project code:**
- **Group name:** Group 2
- **Software Type:** Web application

The rapid development of Information Technology has significantly transformed the way students learn, develop professional skills, and prepare for future careers. In particular, Software Engineering students are required to continuously update their technical knowledge, practical experience, and industry trends to remain competitive in the job market.

However, many students face difficulties in determining a suitable career path, identifying the required technical competencies, evaluating their current skill levels, and planning an effective learning roadmap. They often rely on scattered learning resources, inconsistent career advice, or personal experiences without systematic guidance. Consequently, students may spend considerable time learning technologies that are not aligned with their career goals or current industry demands.

The **Personalized Career Orientation & Learning Roadmap Platform for Software Engineering Students** is proposed to address these challenges. The platform provides intelligent career guidance by analyzing students' academic background, technical skills, career interests, and learning progress. Based on this information, the system generates personalized learning roadmaps, recommends appropriate learning resources, identifies missing competencies through skill gap analysis, and helps students build professional portfolios using their GitHub projects.

By integrating Artificial Intelligence, career planning, learning management, portfolio generation, and job market analysis into a unified platform, the proposed system aims to improve students' learning efficiency and enhance their career readiness.

1.2 Project Team

Table I.1: Project Team

Full Name	Role	Email	Mobile
Nguyen Van Bao	Project Manager		
Phan Thi Duy Chung	Business Analyst		
Vo Thi Kiem Dung	System Analyst		
Tran Thi Bich Diem	UI/UX Designer		
Vo Trong Nguyen	Backend Developer		
Vu Le Hoang Nhat	Frontend Developer		
Vo Bao Phuc	Tester		

2. Product Background

Software Engineering education focuses not only on theoretical knowledge but also on practical skills, project experience, and continuous learning. Modern software industries require graduates to possess diverse competencies, including programming, database management, software architecture, cloud computing, DevOps, artificial intelligence, and communication skills.

Despite the availability of numerous online learning platforms, students often encounter several challenges:

- Lack of personalized career guidance.
- Difficulty selecting an appropriate Software Engineering specialization.
- Limited understanding of industry-required technical skills.
- Inability to evaluate their own competencies objectively.
- Difficulty organizing learning plans according to career objectives.
- Lack of professional portfolios for internship and job applications.

Currently, students usually collect learning materials from various websites such as Microsoft Learn, Coursera, Udemy, roadmap.sh, GitHub, YouTube, and Stack Overflow. Since these resources are independent and not integrated, students must manually determine what to study, when to study, and how to evaluate their progress.

Therefore, a comprehensive platform that combines career orientation, personalized learning roadmap generation, AI-assisted mentoring, skill assessment, portfolio management, and labor market analysis can significantly improve students' learning outcomes and employment opportunities.

3. Existing Systems

Several existing platforms provide partial support for career planning or online learning; however, none of them fully integrate all required functionalities for Software Engineering students.

Roadmap.sh

Roadmap.sh provides structured technology roadmaps for different software engineering career paths, including Backend Development, Frontend Development, DevOps, AI Engineering, and Mobile Development. Although it offers detailed technology trees, it does not personalize learning paths based on students' individual competencies or academic performance.

Coursera

Coursera offers thousands of online courses from universities and technology companies. Students can obtain certificates after completing courses. However, course recommendations are primarily based on browsing history rather than comprehensive career analysis.

LinkedIn Learning

LinkedIn Learning provides professional courses and integrates with LinkedIn profiles. Nevertheless, it lacks personalized skill gap analysis and automatic roadmap generation tailored to Software Engineering students.

GitHub

GitHub serves as the largest source code hosting platform, allowing students to manage software projects and demonstrate programming experience. However, GitHub itself does not evaluate students' technical competencies or generate learning recommendations.

ChatGPT and Gemini

Artificial Intelligence chatbots such as ChatGPT and Gemini can answer programming and career-related questions. Nevertheless, they do not maintain long-term learning progress or automatically generate structured career roadmaps integrated with students' academic profiles.

Therefore, there is still a need for an integrated platform specifically designed for Software Engineering students.

4. Business Opportunity

The software industry continues to experience rapid growth worldwide. Companies increasingly demand graduates with practical technical skills rather than solely academic qualifications.

Universities are also shifting toward outcome-based education, emphasizing competency assessment, project-based learning, and career preparation. Consequently, educational institutions require systems capable of supporting students throughout their

learning journeys.

The proposed platform provides several business opportunities:

- Supporting universities in career counseling.
- Assisting students in planning personalized learning paths.
- Helping employers evaluate candidates through dynamic portfolios.
- Assisting academic advisors in monitoring students' learning progress.
- Providing technology trend analysis based on labor market demands.
- Integrating Artificial Intelligence into educational support systems.

Furthermore, the platform can be expanded in the future to support multiple universities, online certification providers, internship management systems, and enterprise recruitment platforms.

5. Software Product Vision

The vision of the project is to become an intelligent career development platform that accompanies Software Engineering students throughout their academic journey and professional development.

The system aims to:

- Provide personalized career recommendations.
- Generate intelligent learning roadmaps.
- Continuously analyze students' skill gaps.
- Recommend high-quality learning resources.
- Build professional online portfolios automatically.
- Analyze current technology trends and employment demands.
- Improve students' employability and career readiness through data-driven recommendations.

By integrating Artificial Intelligence, GitHub repositories, academic information, and industry trends, the platform creates a comprehensive ecosystem for career development.

6. Project Scope & Limitations

6.1 Major Features

6.1.1 Major Features for Administrators

FE-01: Login/Logout.

FE-02: Manage user accounts (Students, Academic Counselors, Industry Mentors).

FE-03: Manage career paths (Backend Developer, Frontend Developer, Full Stack Developer, DevOps Engineer, AI Engineer, Data Engineer, Mobile Developer, QA Engineer, etc.).

FE-04: Manage skill categories and technical skills.

FE-05: Manage learning resources (courses, books, documentation, videos, practice websites).

FE-06: Manage AI prompt templates and system configurations.

FE-07: Manage job market data and technology trend information.

FE-08: Manage notifications and announcements.

FE-09: View system reports and analytics.

FE-10: Monitor user activities and system logs.

FE-11: Manage role-based access permissions.

6.1.2 Major Features for Student

FE-01: Register, login, logout, and authenticate using Google OAuth.

FE-02: Manage personal profile, including academic information, technical skills, career interests, and learning goals.

FE-03: Complete skill assessment to evaluate current competencies.

FE-04: Receive AI-powered career consultation through the AI Virtual Mentor.

FE-05: Select a target software engineering career path.

FE-06: Generate a personalized learning roadmap based on selected career goals.

FE-07: View and update learning progress for each roadmap milestone.

FE-08: Perform skill gap analysis to identify missing competencies.

FE-09: Receive recommended learning resources based on skill gaps.

FE-10: Connect and synchronize GitHub repositories.

FE-11: Automatically generate and maintain an online professional portfolio.

FE-12: View job market trends, technology demand, and career insights.

FE-13: Receive notifications about roadmap updates, learning recommendations, and career opportunities.

6.1.3 Major Features for Academic Counselor

FE-01: Login/Logout.

FE-02: View student profiles and academic information.

FE-03: Monitor students' learning roadmap progress.

FE-04: Review students' skill gap analysis reports.

FE-05: Provide academic advice and learning recommendations.

FE-06: Track students' learning performance and completion status.

FE-07: Send notifications or reminders to students.

FE-08: View reports summarizing students' learning progress.

6.1.4 Major Features for Industry Mentor

FE-01: Login/Logout.

FE-02: Review students' career roadmaps.

FE-03: Evaluate students' professional portfolios.

FE-04: Recommend industry-required technical skills.

FE-05: Provide career advice and industry feedback.

FE-06: Suggest learning priorities according to current market demand.

FE-07: Review students' project experiences and GitHub repositories.

FE-08: View students' competency development reports.

6.2 Limitations & Exclusions

LI-01: The first version of the system is designed exclusively for Software Engineering students and does not currently support students from other academic disciplines.

LI-02: AI career recommendations depend on external AI services (such as OpenAI or Gemini). Recommendation quality may vary depending on the AI model and available information.

LI-03: GitHub integration supports only repositories that are publicly accessible or explicitly authorized by users.

LI-04: Job market analysis depends on external recruitment platforms and publicly available datasets. Real-time accuracy cannot be fully guaranteed.

LI-05: The system currently supports predefined Software Engineering career paths only. Users cannot create custom career paths.

LI-06: The platform is implemented as a web application. Native Android and iOS applications are not included in this version.

LI-07: The platform requires a stable Internet connection because several core services depend on external APIs (AI services, Google Authentication, GitHub API, and job

market data).

LI-08: The current version does not provide online coding practice, automatic code assessment, or integrated programming environments.

II. Project Management Plan

1. Overview

1.1 Project Scope and Estimation

The **Personalized Career Orientation & Learning Roadmap Platform for Software Engineering Students** is a web-based system designed to assist Software Engineering students in identifying suitable career paths and generating personalized learning roadmaps based on their academic performance, technical skills, interests, and career goals.

The project consists of the following major work packages.

Table II.1: Work Breakdown Structure (WBS) and Project Effort Estimation

WBS	Task	Complexity	Estimated Effort (Man-days)
1	Project Initiation	Medium	10
1.1	Requirement Gathering	Medium	5
1.2	Feasibility Analysis	Medium	5
2	Planning	Medium	8
2.1	Project Planning	Medium	4
2.2	Risk Planning	Medium	4
3	Analysis and Design	Complex	22
3.1	Requirement Analysis	Medium	8
3.2	Database Design	Medium	6
3.3	System Architecture	Complex	8
4	Implementation	Complex	80
4.1	Authentication Module	Simple	6
4.2	Student Profile Module	Medium	8
4.3	Career Assessment Module	Complex	12
4.4	Career Recommendation Engine	Complex	16
4.5	Learning Roadmap Generator	Complex	16
4.6	Learning Resource Management	Medium	10
4.7	Dashboard and Progress Tracking	Medium	12
5	Testing	Complex	24
5.1	Unit Testing	Medium	8
5.2	Integration Testing	Complex	8
5.3	System Testing	Complex	8
6	Deployment	Medium	8
6.1	Deployment	Medium	4
6.2	User Acceptance Testing	Medium	4

Total Estimated Effort: 152 Man-days

The objectives of the project are as follows.

- Develop a web-based personalized career orientation platform.
- Recommend suitable Software Engineering career paths.
- Generate personalized learning roadmaps.
- Track students' learning progress.
- Recommend appropriate learning resources.
- Improve students' career planning capability.
- Complete the project within one academic semester.

1.2 Testing Objectives

Table II.2: Testing Objectives

Testing Stage	Coverage	Target Defects
Code Review	Coding Standard	<10
Unit Testing	Business Logic	<8
Integration Testing	Module Communication	<5
System Testing	Complete Workflow	<5
Acceptance Testing	User Requirements	<3

1.3 Project Risks

Table II.3: Project Risks

Risk	Impact	Probability	Mitigation
Changing Requirements	High	Medium	Weekly meetings with supervisor
Implementation Delay	High	Medium	Sprint planning and monitoring
Incorrect Recommendation Results	Medium	Medium	Validation and testing
Database Performance Issues	Medium	Low	Query optimization
Communication Problems	Medium	Medium	Weekly meetings and GitHub tracking
Member Absence	High	Low	Knowledge sharing and backup assignments

2. Management Approach

2.1 Development Methodology

The project adopts the Agile Scrum methodology.

The reasons include:

- Continuous customer feedback
- Flexible requirement changes
- Incremental software delivery
- Continuous testing
- Better project monitoring

Each sprint lasts two weeks.

2.2 Quality Management

Quality assurance activities include:

- Requirement Review
- Code Review
- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing

2.3 Training Plan

Table II.4: Training Plan

Technology	Participants	Duration
ASP.NET Core	All Members	1 Week
ReactJS	Front-end Developers	1 Week
SQL Server	All Members	3 Days
Git and GitHub	All Members	2 Days
Figma	UI Designer	2 Days

3. Project Deliverables

Table II.5: Project Deliverables

Sprint	Deliverables
1	Requirement Analysis
2	Database Design and UI Prototype
3	Authentication Module
4	Student Profile Module
5	Career Assessment Module
6	Career Recommendation Module
7	Learning Roadmap Module
8	Learning Resource Module
9	Dashboard and Progress Tracking
10	Integration Testing
11	System Testing
12	Deployment and Final Report

4. Responsibility Assignments

Table II.6: Responsibility Assignment Matrix

Task	Member 1	Member 2	Member 3	Member 4
Requirement Analysis	D	R	S	S
Project Planning	D	S	R	S
Database Design	R	D	S	S
Backend Development	S	D	R	S
Frontend Development	S	R	D	S
Testing	R	S	D	S
Documentation	S	R	S	D

Where

- D = Do
- R = Review
- S = Support

5. Project Communications

Table II.7: Project Communications

Communication	Participants	Frequency	Tool
Daily Meeting	Development Team	Daily	Google Meet
Sprint Planning	All Members	Every 2 Weeks	Google Meet
Sprint Review	Supervisor	Every Sprint	Offline/Meet
Issue Discussion	Related Members	As Needed	Discord
Document Sharing	All Members	Continuous	Google Drive
Source Code Review	Developers	Continuous	GitHub

6. Configuration Management

6.1 Document Management

The project documentation is maintained using:

- Microsoft Word
- Microsoft Excel
- Google Drive

6.2 Source Code Management

The source code is managed using GitHub and Git.

The following branching strategy is adopted:

- main
- develop
- feature/*
- release/*
- hotfix/*

6.3 Version Control Policy

- Every feature is developed in an independent branch.
- Pull Requests are mandatory before merging.

- Code review is required before approval.
- Semantic Versioning is applied.

Example versions:

- Version 1.0.0
- Version 1.1.0
- Version 2.0.0

III. Software Requirement Specification

1. Product Overview

2. User Requirements

3. Functional Requirements

3.1 System Functional Overview

3.2 Authentication

Authentication is responsible for verifying user identities before allowing access to the Personalized Career Orientation & Learning Roadmap Platform. This module provides secure login, user registration, password recovery, and logout functions. Only authenticated users can access system resources according to their assigned roles.

3.2.1 Login

Role:

Student, Mentor, Career Counselor, Administrator.

Function Trigger:

The user enters a registered email address and password, then clicks the **Login** button.

Function Description:

- **Purpose:**

Authenticate users before granting access to the system.

- **Data Processing:**

The system validates the entered credentials, verifies the account status, authenticates the user, determines the user role, and redirects the user to the corresponding dashboard.

Function Details:

- If the login credentials are correct, the system redirects the user to the corresponding dashboard.
- If the email address or password is incorrect, the system displays an error message.
- If the account has not been verified, the system requests email verification before login.
- If the account is temporarily locked because of multiple unsuccessful login attempts, the system informs the user accordingly.



SE Career Compass
Roadmap platform

STUDENT PORTAL

Sign in to continue

Access your roadmap, skill gap report, mentor notes, and portfolio records.

Sign in

Create account

Email

Password

☒ Keep me signed in on this device

→] Sign in

 Continue with Google

Figure III.1: Login Interface

3.3 User Management

User Management is responsible for managing user accounts and personal information within the Personalized Career Orientation & Learning Roadmap Platform. This module enables administrators to maintain user accounts, while users can update their own profile information. The system also supports searching, viewing, editing, and deleting

user accounts according to role-based access permissions.

3.3.1 Manage Users

Role:

Administrator.

Function Trigger:

The administrator logs into the system and navigates to the **User Management** page.

Function Description:

- **Purpose:**
Manage all user accounts in the system.
- **Data Processing:**
Create, edit, delete, search, and update user account information.

Function Details:

- The administrator can create new user accounts.
- The administrator can update user information.
- The administrator can activate or deactivate user accounts.
- Search users by Student ID, Full Name, or Email Address.
- Assign roles such as Student, Mentor, Career Counselor, or Administrator.

3.3.2 Import Students

Role:

Administrator.

Function Trigger:

The administrator logs into the system, opens the **User Management** module, and selects the **Import Students** function.

Function Description:

- **Purpose:**
Import student accounts into the system.

- **Data Processing:**

Read student information from an Excel file and automatically create user accounts.

Function Details:

- Import student information from an Excel file.
- Validate duplicated Student IDs and email addresses.
- Automatically assign the Student role.
- Display the import result after processing.

3.3.3 Import Mentors

Role:

Administrator.

Function Trigger:

The administrator logs into the system, opens the **User Management** module, and selects the **Import Mentors** function.

Function Description:

- **Purpose:**

Import mentor accounts into the system.

- **Data Processing:**

Read mentor information from an Excel file and create mentor accounts automatically.

Function Details:

- Import mentor information from an Excel file.
- Validate duplicated email addresses.
- Automatically assign the Mentor role.
- Display the import result after processing.

3.4 Career Assessment Management

Career Assessment Management allows students to evaluate their technical skills, interests, and career orientation through assessment questionnaires. The assessment results are analyzed by the system and used as the basis for generating personalized career recommendations and learning roadmaps.

3.4.1 Student Take Career Assessment

Function Trigger:

Student logs in → navigates to "**Career Assessment**" → clicks "**Start Assessment**".

Function Description:**Actors:**

Student

Purpose:

Allow students to evaluate their technical skills, interests, and career orientation.

Interface:

An assessment page containing multiple-choice questions related to programming, software engineering knowledge, career interests, and technical skills.

Data Processing:

The student's answers are saved in the database. The system calculates the assessment score and generates an assessment result for future AI recommendations.

Function Details:**Validation:**

- All questions must be answered.

Business Rules:

- Students can take the assessment once per evaluation period.
- Assessment results are stored for future comparison.

Normal Case:

Student answers all questions → clicks "**Submit**" → system calculates the score and saves the result.

Abnormal Case:

If required questions are unanswered, the system requests the student to complete all questions before submission.

3.4.2 View Assessment Result

Function Trigger:

Student completes the assessment → selects "**View Result**".

Function Description:

Actors:

Student

Purpose:

Allow students to review their assessment results.

Interface:

A result page displaying assessment score, strengths, weaknesses, and suggested career paths.

Data Processing:

Retrieve assessment results from the database and display them graphically.

Function Details:**Business Rules:**

- Students can only view their own assessment results.

Normal Case:

Student selects "**View Result**" → system displays assessment report.

Abnormal Case:

If no assessment exists, the system requests the student to complete an assessment first.

3.4.3 Mentor Review Assessment

Function Trigger:

Mentor logs in → navigates to "**Student Assessments**" → selects a student.

Function Description:**Actors:**

Mentor

Purpose:

Review students' assessment results and provide career guidance.

Interface:

A dashboard displaying assessment reports, skill analysis, and feedback form.

Data Processing:

Store mentor comments and recommendations in the database.

Function Details:

Business Rules:

- Mentors can only review students assigned to them.

Normal Case:

Mentor reviews assessment → enters comments → clicks "**Save Feedback**".

Abnormal Case:

If the selected student has not completed an assessment, the system displays a notification.

3.4.4 Manage Assessment Questions

Function Trigger:

Administrator logs in → navigates to "**Assessment Management**".

Function Description:**Actors:**

Administrator

Purpose:

Manage assessment questions used by the career assessment module.

Interface:

A management page allowing administrators to create, edit, delete, and organize assessment questions.

Data Processing:

Save updated assessment questions to the database.

Function Details:**Business Rules:**

- Only administrators can modify assessment questions.

Normal Case:

Administrator creates or updates questions → system saves the changes.

Abnormal Case:

If required fields are missing, the system displays validation messages.

3.5 Mentorship Management

Mentorship Management allows students to connect with mentors for career guidance and learning support. Students can request mentorship, while mentors can review,

accept, or decline requests. The system also enables both parties to manage mentorship relationships throughout the learning process.

3.5.1 Student Request Mentor

Function Trigger:

Student logs in → navigates to **"Find Mentors"** → selects a mentor → clicks **"Request Mentorship"**.

Function Description:

Actors:

Student

Purpose:

Allow students to request mentorship from a mentor.

Interface:

A mentor profile page displaying mentor information, expertise, availability, and a "Request Mentorship" button.

Data Processing:

The system creates a mentorship request with the status **Pending** and notifies the selected mentor.

Function Details:

Validation:

All required fields must be completed.

Business Rules:

A student can only have one active mentor at a time.

A mentorship request cannot be duplicated.

Normal Case:

Student submits a mentorship request → system stores the request and displays a success message.

Abnormal Case:

Duplicate request or mentor unavailable → display an error message.

3.5.2 Mentor Review Request

Function Trigger:

Mentor logs in → opens **"Mentorship Requests"**.

Function Description:**Actors:**

Mentor

Purpose:

Allow mentors to review mentorship requests.

Interface:

A list of pending mentorship requests with student information and request details.

Data Processing:

The system retrieves pending requests and displays them for review.

Function Details:

Mentors can review student profiles before making a decision.

3.5.3 Mentor Respond Request**Function Trigger:**

Mentor selects a mentorship request → clicks "**Accept**" or "**Reject**".

Function Description:**Actors:**

Mentor

Purpose:

Approve or reject a student's mentorship request.

Interface:

A mentorship request detail page with Accept and Reject buttons.

Data Processing:

Update the request status and notify the student.

Function Details:**Business Rules:**

A mentor cannot exceed the maximum number of active mentees.

Normal Case:

Mentor accepts the request → mentorship relationship is created.

Abnormal Case:

Maximum mentee limit reached → display an error message.

3.5.4 Schedule Mentoring Session

Function Trigger:

Student or Mentor logs in → selects "Schedule Session".

Function Description:

Actors:

Student, Mentor

Purpose:

Schedule mentoring sessions between students and mentors.

Interface:

A scheduling calendar with available dates and meeting information.

Data Processing:

Save meeting information and notify both participants.

Function Details:

Meeting schedules are synchronized for both users.

3.5.5 View Mentoring History

Function Trigger:

Student or Mentor logs in → opens "Mentoring History".

Function Description:

Actors:

Student, Mentor

Purpose:

Display previous mentoring sessions and recommendations.

Interface:

A table showing meeting history, mentor feedback, and follow-up recommendations.

Data Processing:

Retrieve mentoring records from the database.

Function Details:

Users can search and filter mentoring history by date.

3.6 Learning Roadmap Management

Learning Roadmap Management enables students to generate, view, and update personalized learning roadmaps based on their career goals, current skills, and AI recommendations. The module helps students monitor their learning progress and achieve the competencies required for their desired career paths.

3.6.1 Generate Learning Roadmap

Function Trigger:

Student logs in → navigates to "**Learning Roadmap**" → clicks "**Generate Roadmap**".

Function Description:

Actors:

Student

Purpose:

Generate a personalized learning roadmap based on the student's assessment results and career objectives.

Interface:

A page allowing students to select a career path, preferred learning duration, and learning pace before generating the roadmap.

Data Processing:

The system analyzes assessment results and AI recommendations to generate a customized learning roadmap, which is then stored in the database.

Function Details:

Validation:

The student must complete the career assessment before generating a roadmap.

Business Rules:

Each student can maintain only one active learning roadmap for each selected career path.

Normal Case:

Student selects career preferences → clicks "**Generate Roadmap**" → the system creates and displays the personalized roadmap.

Abnormal Case:

If the student has not completed the assessment, the system requests completion before generating the roadmap.

3.6.2 Update Learning Progress

Function Trigger:

Student logs in → opens the learning roadmap → marks a learning activity as completed.

Function Description:**Actors:**

Student

Purpose:

Allow students to update their learning progress.

Interface:

A roadmap page displaying learning milestones with progress indicators and completion checkboxes.

Data Processing:

The system updates the student's learning progress and recalculates the completion percentage.

Function Details:**Validation:**

Only authenticated students can update their own learning progress.

Business Rules:

Completed learning activities cannot be modified unless reset by the student.

Normal Case:

Student marks a task as completed → system updates the progress successfully.

Abnormal Case:

If the selected task does not exist, the system displays an error message.

3.6.3 Export Learning Roadmap

Function Trigger:

Student logs in → opens the learning roadmap → clicks "**Export**".

Function Description:**Actors:**

Student

Purpose:

Allow students to export their personalized learning roadmap for future reference.

Interface:

An export dialog allowing students to select PDF or Excel format.

Data Processing:

The system generates the selected document format and downloads it to the user's device.

Function Details:

Validation:

The student must have an existing learning roadmap before exporting.

Business Rules:

Only the owner of the roadmap can export it.

Normal Case:

Student selects the export format → clicks "**Download**" → system generates and downloads the file.

Abnormal Case:

If no learning roadmap exists, the system displays an appropriate notification.

3.7 Learning Resource Management

Learning Resource Management enables students to discover, access, and manage learning resources recommended by the system. These resources include online courses, books, tutorials, technical documentation, and certification programs that support students in achieving their career goals.

3.7.1 Generate Learning Resource Recommendations

Function Trigger:

Student logs in → navigates to "**Learning Resources**" → selects a career path or learning roadmap.

Function Description:

Actors:

Student

Purpose:

Generate personalized learning resource recommendations based on the student's career objectives, assessment results, and learning roadmap.

Interface:

A recommendation page displaying filters for career path, technology, difficulty level, and learning platform.

Data Processing:

The system analyzes the student's profile and retrieves suitable learning resources from the database and external learning platforms.

Function Details:

Validation:

The student must complete a career assessment before receiving personalized recommendations.

Business Rules:

Only active and verified learning resources are recommended.

The recommendation list is updated whenever the student's learning roadmap changes.

Normal Case:

Student selects a career path → system generates recommended learning resources → recommendation list is displayed successfully.

Abnormal Case:

If no suitable learning resources are found, the system displays an appropriate notification.

3.7.2 Manage Learning Resources

Function Trigger:

Administrator logs in → navigates to "**Learning Resource Management**".

Function Description:

Actors:

Administrator

Purpose:

Manage learning resources available on the platform.

Interface:

A management page allowing administrators to add, edit, delete, search, and categorize learning resources.

Data Processing:

The system validates the entered information and updates the learning resource database.

Function Details:

Validation:

All required fields must be completed.

Resource links must be valid URLs.

Business Rules:

Only administrators can modify learning resources.

Duplicate learning resources are not allowed.

Normal Case:

Administrator enters resource information → clicks "**Save**" → the system stores the resource successfully.

Abnormal Case:

If required information is missing or the resource already exists, the system displays an error message.

3.8 Timeline Management

The Timeline Management module allows administrators to define and manage important milestones throughout students' career development process. It enables administrators to configure schedules for career assessments, AI recommendation periods, mentoring sessions, learning roadmap updates, portfolio reviews, and career consultation activities. Proper timeline management ensures that all users receive notifications and complete required activities according to the academic calendar.

3.8.1 Manage Career Development Timeline

Function Trigger:

Administrator logs in → navigates to "Timeline Management" → selects an academic semester or training period.

Function Description:

Actors:

Administrator

Purpose:

Allow administrators to create, update, and manage important milestones for career orientation, AI assessments, mentoring activities, and learning roadmap execution.

Interface:

The interface contains:

- Academic Semester
- Career Assessment Period
- AI Recommendation Schedule

- Learning Roadmap Update Period
- Mentor Consultation Schedule
- Portfolio Submission Deadline
- Career Fair Schedule
- Save button
- Cancel button

Data Processing:

The system retrieves the existing academic timeline from the database. After validation, updated schedules are stored and notifications are automatically sent to affected users.

Function Details:

Validation:

- All required timeline information must be completed.
- Dates must follow chronological order.
- Start dates must be earlier than end dates.
- Activities cannot overlap if conflicts exist.

Business Rules:

- Each semester contains only one official career development timeline.
- Career assessment must be completed before AI recommendation generation.
- Learning roadmap generation is available only after assessment completion.
- Mentor consultation sessions must occur after learning roadmap generation.
- Portfolio evaluation takes place near the end of the semester.
- All timeline updates are recorded in the audit log.

Normal Case:

1. Administrator opens Timeline Management.
2. Selects an academic semester.
3. Updates important schedules.

4. Clicks **Save**.
5. The system validates all dates.
6. Timeline information is saved successfully.
7. Notifications are sent to related users.
8. The system displays:
 “Timeline updated successfully.”

Abnormal Case:

- Invalid date format.
 “Please enter a valid date.”
- End date is earlier than start date.
 “End date must be after start date.”
- Timeline conflicts with existing activities.
 “Timeline conflict detected.”
- Database error.
 “Unable to update the timeline. Please try again later.”

3.9 Blog Management

The Blog Management module allows students, mentors, and career counselors to share knowledge, learning experiences, career guidance, technology trends, internship opportunities, and other software engineering related content. Users can create blog posts, interact with published articles, and participate in knowledge sharing within the platform community.

3.9.1 Create Blog Post

Function Trigger:

Student or Mentor logs in → navigates to “Blog” → clicks the **Create New Post** button.

Function Description:

Actors:

Student, Mentor

Purpose:

Allow authorized users to publish articles related to career orientation, learning experiences, technical knowledge, internship opportunities, software engineering technologies, and academic activities.

Interface:

The blog creation page contains:

- Blog Title
- Blog Category
- Rich Text Editor
- Upload Cover Image
- Upload Attachment (Optional)
- Publish Button
- Save Draft Button

Data Processing:

The system validates all submitted information before saving the blog post into the database. If the post is published successfully, it is displayed on the Blog page and notifications may be sent to followers or subscribed users.

Function Details:**Validation:**

- Blog title is required.
- Blog content is required.
- Uploaded images must be JPG, JPEG or PNG format.
- Maximum image size is 5 MB.

Business Rules:

- Only authenticated students and mentors can create blog posts.
- Blog posts may be reviewed by administrators if reported.
- Draft posts are only visible to their authors.
- Published posts become immediately visible to all users.

Normal Case:

1. User opens the Blog module.
2. Clicks **Create New Post**.
3. Enters the title and content.
4. Uploads an optional cover image.
5. Clicks **Publish**.
6. The system validates all information.
7. The blog post is stored successfully.
8. The post appears on the Blog page.

Abnormal Case:

- Missing title.
“Please enter the blog title.”
- Missing content.
“Please enter blog content.”
- Unsupported image format.
“Only JPG, JPEG or PNG images are accepted.”
- Uploaded image exceeds the maximum size.
“Maximum image size is 5 MB.”
- System error.
“Unable to publish the blog post. Please try again later.”

3.9.2 Interact with Blog Post

Function Trigger:

User logs in → navigates to the “Blog” page → selects a blog post → clicks the **Like** button or submits a comment.

Function Description:

Actors:

Student, Mentor, Career Counselor

Purpose:

Allow users to interact with blog posts by expressing appreciation through likes and participating in discussions by posting comments.

Interface:

The blog detail page contains:

- Blog content
- Like button
- Total number of likes
- Comment section
- Comment text box
- Submit Comment button

Data Processing:

The system records user interactions in the database. Likes are counted immediately, while comments are stored and displayed below the corresponding blog post. Notifications may be sent to the blog author when new interactions occur.

Function Details:

Validation:

- Comments cannot be empty.
- The selected blog post must exist.
- Only authenticated users can interact with blog posts.

Business Rules:

- Each user can like a blog post only once.
- Clicking the Like button again removes the previous like.
- Comments containing inappropriate language may be automatically filtered.
- Deleted blog posts cannot receive new likes or comments.

Normal Case:

1. User opens a blog post.
2. Clicks the Like button or enters a comment.
3. Clicks Submit Comment.
4. The system validates the submitted information.
5. The interaction is saved successfully.
6. The updated number of likes or comments is displayed immediately.

Abnormal Case:

- Empty comment.
“Please enter your comment.”
- The selected blog post no longer exists.
“This blog post is no longer available.”
- Network or database error.
“Unable to process your request. Please try again later.”

3.9.3 Apply for Internship or Team Recruitment**Function Trigger:**

Student logs in → navigates to the “Blog” page → opens an internship or team recruitment post → clicks the **Apply** button.

Function Description:**Actors:**

Student

Purpose:

Allow students to apply for internship opportunities, project teams, research groups, or mentoring programs by submitting their CV and optional cover letter.

Interface:

The application page contains:

- Recruitment Information
- Upload CV
- Cover Letter
- Apply Button
- Cancel Button

Data Processing:

The system validates the uploaded CV, stores the application in the database, links the application to the corresponding recruitment post, and notifies the recruiter or mentor.

Function Details:**Validation:**

- CV must be uploaded.

- Accepted file formats are PDF and DOCX.
- Maximum file size is 5 MB.
- Cover Letter is optional but cannot exceed 1000 characters.

Business Rules:

- A student may submit only one application for each recruitment post.
- Applications can be updated only before the recruitment deadline.
- Students must have a completed personal profile before submitting an application.
- Recruiters receive notifications immediately after a successful application.

Normal Case:

1. Student opens a recruitment blog post.
2. Uploads a valid CV.
3. Enters an optional cover letter.
4. Clicks the Apply button.
5. The system validates the submitted information.
6. The application is stored successfully.
7. A confirmation message is displayed.
8. The recruiter receives a notification.

Abnormal Case:

- No CV is uploaded.
“Please upload your CV.”
- Unsupported file format.
“Only PDF and DOCX files are accepted.”
- Uploaded file exceeds the maximum size.
“The maximum allowed file size is 5 MB.”
- The student has already applied.
“You have already applied for this opportunity.”
- System error.
“Unable to submit your application. Please try again later.”

3.10 Semester Management

The Semester Management module allows administrators or managers to create, configure, and maintain academic semesters used throughout the system. Each semester contains important information such as the semester period, review schedules, defense schedules, and other academic milestones. Proper semester management ensures that all project activities are organized according to the official academic timeline.

3.10.1 Manage Semester

Role:

Administrator, Manager.

Function Trigger:

The Manager or Administrator logs into the system, navigates to the **Semester Management** module, and selects the **Manage Semester** function.

Function Description:

Purpose:

Allow authorized users to create, edit, update, and manage semester information, including review schedules, defense schedules, and important academic milestones.

Interface:

The system displays a semester management page containing:

- Semester list.
- Add Semester button.
- Edit Semester button.
- Delete Semester button.
- Semester information form including:
 - Semester Name
 - Start Date
 - End Date
 - Review Schedule
 - Defense Schedule
 - Additional Notes

Data Processing:

The system validates all entered information before saving it into the Semester database. After successful validation, the system updates the semester information and synchronizes related review and defense schedules.

Function Details:

Validation:

- All required fields must be completed.
- Semester dates must follow the correct date format.
- Review and defense schedules must fall within the semester duration.
- Semester periods must not overlap with existing semesters.

Business Rules:

- Only Managers and Administrators are allowed to manage semesters.
- Every semester must contain exactly one academic timeline.
- Review sessions must be scheduled before defense sessions.
- All schedule changes are recorded in the system log.

Normal Case:

1. The Manager opens the Semester Management page.
2. The Manager enters or edits semester information.
3. The Manager clicks the **Save** button.
4. The system validates all information.
5. The semester information is stored successfully.
6. The system displays the message:

"Semester information has been updated successfully."

Abnormal Case:

- Missing required information.
- Invalid date format.
- Semester dates overlap with an existing semester.
- Review or defense schedules are outside the semester period.
- Unexpected system error during data saving.

3.11 Criteria Form Management

The Criteria Form Management module enables managers to create, modify, and maintain evaluation forms and assessment criteria used throughout the project lifecycle. These forms are applied during review and defense sessions to ensure that all evaluations follow standardized criteria and grading rules.

3.11.1 Manage Criteria Form

Role:

Manager.

Function Trigger:

The Manager logs into the system, navigates to the **Form Management** module, and selects the **Add/Edit Form, Criteria** function.

Function Description:

Purpose:

Allow managers to create, update, and manage evaluation forms and assessment criteria used for reviews and defense sessions in each semester.

Interface:

The system displays a management page containing:

- List of existing evaluation forms.
- Add Form button.
- Edit Form button.
- Delete Form button.
- Form information including:
 - Form Title
 - Applicable Semester
 - Evaluation Criteria
 - Question
 - Score Type
 - Weight
 - Description

Data Processing:

The system validates all entered information before storing it in the Form and Criteria database tables. Each form is associated with a semester and can later be reused in future semesters if required.

Function Details:

Validation:

- All mandatory fields must be completed.
- The form title must be unique within the same semester.
- Each criterion must contain a valid question and score type.
- Mandatory criteria cannot be left empty.
- The applicable semester must exist in the system.

Business Rules:

- Only Managers are allowed to create, edit, or delete evaluation forms.
- Evaluation forms may be reused in multiple semesters.
- Mandatory criteria must always be included in every evaluation form.
- Lecturers are authorized to evaluate students using the assigned forms.
- Council members are authorized to evaluate lecturers during defense sessions.

Normal Case:

1. The Manager opens the Form Management page.
2. The Manager creates or edits an evaluation form.
3. The Manager enters all required criteria.
4. The Manager clicks the **Save** button.
5. The system validates all information.
6. The evaluation form is stored successfully.
7. The system displays the message:
"Evaluation form has been saved successfully."

Abnormal Case:

- Required information is missing.

- Duplicate form title within the same semester.
- Invalid criterion configuration.
- Invalid semester selection.
- Unexpected system error during data saving.

4. Non-Functional Requirements

This section describes the non-functional requirements of the Personalized Career Orientation & Learning Roadmap Platform. These requirements specify the quality characteristics that the system shall satisfy, including external interfaces, usability, reliability, and performance.

4.1 External Interfaces

To ensure effective communication between users and external software services, the system shall provide the following interfaces.

User Interface

- The interface shall be intuitive, user-friendly, and easy to use for Students, Mentors, Career Counselors, and Administrators.
- The interface shall follow modern GUI design standards to reduce user learning time.
- The platform shall support desktop computers, tablets, and smartphones through responsive web design.

Artificial Intelligence Interface

- The system shall integrate with Artificial Intelligence services to generate personalized career recommendations, learning roadmaps, and skill gap analysis.
- Communication between the platform and AI services shall use secure REST APIs over HTTPS.

Authentication Interface

- The system shall support Google Authentication using OAuth 2.0.
- User authentication data shall be transmitted securely using HTTPS.

External Learning Platform Interface

- The platform shall integrate with online learning platforms such as Coursera, Udemy, edX, GitHub Learning Lab, and other educational services.
- Learning resources shall be synchronized whenever external APIs are available.

4.2 Quality Attributes

The platform shall satisfy quality requirements related to usability, reliability, and performance.

4.2.1 Usability

User-Friendly Interface

- The system shall provide an intuitive interface for all user roles.
- Navigation shall remain consistent across all modules.
- Frequently used functions shall be accessible within a few clicks.

Training and Learning Time

- Students shall become familiar with the basic functions within thirty minutes.
- Mentors and administrators shall require no more than two hours to master management functions.

Usability Metrics

- Login shall require less than one minute.
- Users shall access AI career recommendations within three clicks.
- Overall user satisfaction should exceed 90%.

4.2.2 Reliability

Availability

- The system shall maintain at least 99.9% annual availability.
- Scheduled maintenance shall be performed during off-peak hours.

System Recovery

- Critical failures shall be recovered within thirty minutes.
- Automatic database backup shall be performed regularly.

Data Integrity

- Student profiles, learning progress, portfolios, and project information shall remain consistent.

- Unauthorized modification of data shall be prevented.

Error Handling

- System errors shall be recorded automatically.
- Users shall receive clear error messages whenever failures occur.

4.2.3 Performance

Response Time

- Standard page requests shall respond within two seconds.
- AI recommendation generation shall not exceed five seconds.

Throughput

- The platform shall process at least 1,000 requests per minute during peak usage.

Scalability

- The system shall support at least 5,000 concurrent users without significant performance degradation.
- Future modules and external services shall be integrated without major architectural changes.

Resource Management

- CPU, memory, network bandwidth, and database resources shall be continuously monitored.
- Automatic alerts shall be generated when system resources exceed predefined thresholds.

5. Requirement Appendix

This section summarizes the business rules, common system requirements, application messages, and additional requirements that support the functional and non-functional specifications described in previous sections.

5.1 Business Rules

Table III.1: Business Rules

ID	Rule Definition
BR-01	Each student can create only one career profile in the system.
BR-02	A student can follow only one primary learning roadmap at a time.
BR-03	Students must complete the skill assessment before receiving AI career recommendations.
BR-04	Each learning roadmap is generated based on the student's skills, interests, and career goals.
BR-05	Students can upload multiple portfolio projects, but each project belongs to only one student.
BR-06	Mentors can review only the students assigned to them.
BR-07	Administrators have full permission to manage users, learning resources, and career paths.
BR-08	Every major system activity shall be recorded in the system log.
BR-09	Learning progress is automatically updated after a student completes a learning activity.
BR-10	AI recommendations shall be regenerated whenever a student's assessment results are updated.
BR-11	External learning resources must be retrieved only from approved learning platforms.
BR-12	Only authenticated users can access personalized services.

5.2 Common Requirements

Authentication and Authorization

The system shall implement Role-Based Access Control (RBAC). Users may only access the functions permitted by their assigned roles.

Data Privacy

Student profiles, assessment results, learning progress, and portfolios shall be protected using secure authentication and encrypted communication.

Scalability

The platform shall support increasing numbers of users, learning resources, and AI recommendation requests without significant performance degradation.

Availability

The system should remain available for students and mentors throughout the academic semester except during scheduled maintenance periods.

Maintainability

System modules shall be designed independently to simplify future maintenance and

feature enhancement.

5.3 Application Messages List

Table III.2: Application Messages

Code	Type	Context	Message
MSG01	Error	Login	Invalid email address or password.
MSG02	Success	Login	Login successful. Welcome back!
MSG03	Error	Skill Assessment	Please complete all required assessment questions.
MSG04	Success	Learning Roadmap	Your personalized learning roadmap has been generated successfully.
MSG05	Success	Portfolio	Portfolio uploaded successfully.
MSG06	Error	Portfolio	Invalid file format. Please upload PDF, DOCX, or ZIP files only.
MSG07	Info	AI Recommendation	Generating personalized recommendations. Please wait...
MSG08	Success	Profile	Profile updated successfully.
MSG09	Error	Learning Resources	Unable to retrieve learning resources. Please try again later.
MSG10	Success	Mentor Feedback	Feedback submitted successfully.

5.4 Other Requirements

- The system shall support modern web browsers including Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.
- The user interface shall be responsive and compatible with desktop computers, tablets, and smartphones.
- All user activities shall be recorded for auditing and security purposes.
- Passwords shall be encrypted before being stored in the database.
- The system shall perform automatic daily database backups.
- AI recommendation services should remain available whenever external AI services are accessible.
- Future releases may integrate internship recommendation, job matching, online certification providers, and university information systems.